

교양수학 수학I 퀴즈-6 모범답안

#1

sol)

(i) $x+1 = \sec \theta$ ($0 \leq \theta \leq \frac{\pi}{2}$) 로 치환하자.

양변을 x 에 대하여 미분하면 $1 = \sec \theta \tan \theta \frac{d\theta}{dx}$

이므로 $dx = \sec \theta \tan \theta d\theta$ 이다.

$$\int \sqrt{x^2+2x} dx = \int \sqrt{(x+1)^2-1} dx$$

$$= \int \sqrt{\sec^2 \theta - 1} \sec \theta \tan \theta d\theta$$

$$= \int \sqrt{\tan^2 \theta} \sec \theta \tan \theta d\theta \quad (\because \sec^2 \theta = 1 + \tan^2 \theta)$$

$$= \int \sec \theta \tan^2 \theta d\theta \quad (\because 0 \leq \theta \leq \frac{\pi}{2} \text{에서 } \tan \theta > 0)$$

$$= \int \sec \theta (\sec^2 \theta - 1) d\theta$$

$$= \int (\sec^3 \theta - \sec \theta) d\theta$$

$$= \int \sec^3 \theta d\theta - \int \sec \theta d\theta$$

$$= \left(\frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| \right) - \ln |\sec \theta + \tan \theta| + C$$

$$= \frac{1}{2} \sec \theta \tan \theta - \frac{1}{2} \ln |\sec \theta + \tan \theta| + C \quad \dots \textcircled{A}$$

(ii) 변수 θ 를 x 에 대하여 표현하자.

(i)에서 $x+1 = \sec \theta$ ($0 \leq \theta \leq \frac{\pi}{2}$) 이므로

$$\tan^2 \theta = \sec^2 \theta - 1 = (x+1)^2 - 1 = x^2 + 2x$$

$$\tan \theta = \sqrt{x^2 + 2x} \quad \dots \textcircled{B}$$

따라서, ㉠, ㉡에서

$$\begin{aligned}\Theta &= \frac{1}{2} \sec\theta \tan\theta - \frac{1}{2} \ln |\sec\theta + \tan\theta| + C \\ &= \frac{1}{2} (x+1) \sqrt{x^2+2x} - \frac{1}{2} \ln |(x+1) + \sqrt{x^2+2x}| + C\end{aligned}$$

$$\therefore \int \sqrt{2x+x^2} dx = \frac{1}{2} (x+1) \sqrt{x^2+2x} - \frac{1}{2} \ln |(x+1) + \sqrt{x^2+2x}| + C$$

#2

sol) $\int \frac{10}{(x-1)(x^2-9)} dx$

$$= \int \frac{10}{(x-1)(x-3)(x+3)} dx = \int \left\{ \frac{A}{x-1} + \frac{B}{x-3} + \frac{C}{x+3} \right\} dx$$

by ㉠
 $\downarrow$
 $= \int \left\{ -\frac{5}{4} \cdot \frac{1}{x-1} + \frac{5}{6} \frac{1}{x-3} + \frac{5}{12} \cdot \frac{1}{x+3} \right\} dx$

$$= -\frac{5}{4} \ln|x-1| + \frac{5}{6} \ln|x-3| + \frac{5}{12} \ln|x+3| + C$$

$$\begin{aligned}\textcircled{7}: \frac{10}{(x-1)(x-3)(x+3)} &= \frac{A}{x-1} + \frac{B}{x-3} + \frac{C}{x+3} \\ &= \frac{A(x-3)(x+3) + B(x-1)(x+3) + C(x-1)(x-3)}{(x-1)(x-3)(x+3)}\end{aligned}$$

$$\Leftrightarrow 10 = A(x-3)(x+3) + B(x-1)(x+3) + C(x-1)(x-3)$$

$$x=1 \text{ 대입 : } 10 = -8A \quad \therefore A = -\frac{5}{4}$$

$$x=3 \text{ 대입 : } 10 = 12B \quad \therefore B = \frac{5}{6}$$

$$x=-3 \text{ 대입 : } 10 = 24C \quad \therefore C = \frac{5}{12}$$